3-26-26 (WEEK 5)

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EXAMPLE (RECALL)

Three investment opportunities:

Optn 1: Guaranteed \$1 a year

Optn 2: Get \$1000 w/p $\frac{1}{1000}$ (with prob), \$0 w/p $\frac{999}{1000}$

Optn 3: We get \$$\frac{2000}{999}$ w/p $\frac{999}{1000}$, \$-1000 w/p $\frac{1}{1000}$

What optn should you go for?

Let X_i be the random variable representing the earning of opt i after a year.

Then, $\mathbb{E}[X_1] = \$1 \cdot 1 = 1$

$\mathbb{E}[X_2] = \$1000 \cdot \frac{1}{1000} + \$0 \cdot \frac{999}{1000} = \$1$

$\mathbb{E}[X_3] = \$\frac{2000}{999} \cdot \frac{999}{1000} - \$1000 \cdot \frac{1}{1000} = \$1$

The EVs are the same but clearly they are diff optns and X_2, X_3 don't really resemble their EV

VARIANCE

Variance measures how far one typically is from its expectation

DEFINITION

Let X be a random variable on a probability space $(S, \mathbb{P})$, and let $\mu = \mathbb{E}[X]$

We define the variance to be $\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$

EXAMPLE

$\mathrm{Var}(X_1) = \mathbb{E}[(X_1-1)^2] = (1-1)^2 \cdot 1 = \boxed{0}$

$\mathrm{Var}(X_2) = \mathbb{E}[(X_2-1)^2] = (1000-1)^2 \cdot \frac{1}{1000} + (0-1)^2 \cdot \frac{999}{1000} \approx \boxed{999}$ ← much larger than $X_1 \Rightarrow$ volatile

$\mathrm{Var}(X_3) = \mathbb{E}[(X_3-1)^2] = (\frac{2000}{999}-1)^2 \cdot \frac{999}{1000} + (-1000-1)^2 \cdot \frac{1}{1000} \approx 1003$

CLAIM

For any random variable, $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

<u>Proof</u>

By def, $\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$

$= \mathbb{E}[X^2 - 2X\mu + \mu^2]$

$= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mu + \mu^2$ (linearity of expectation)

$= \mathbb{E}[X^2] - 2\mathbb{E}[X]^2 + \mathbb{E}[X]^2$

$= \mathbb{E}[X^2] - \mathbb{E}[X]^2$ □

COROLLARY

$\forall X$, $\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$ since $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 \geq 0$

REMARK

Sometimes, it is easier to compute $\mathbb{E}[X(X-1)] \overset{\text{by linearity}}{=} \mathbb{E}[X^2] - \mathbb{E}[X]$

$\Rightarrow \mathrm{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2$

DEFINITION

The standard deviation of a random variable X, denoted as $\sigma(X) = \sqrt{\mathrm{Var}(X)}$

(We can sometimes denote variance as $\sigma^2(X)$)

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PROPOSITION (MARKOV'S INEQUALITY)

Let $X \geq 0$ be a nonnegative random variable on a probability space $(S, \mathbb{P})$, and let $a > 0$ be a positive real number

Then, $\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$

Proof

By def, $\mathbb{E}[X] = \sum_x x\,\mathbb{P}(X=x) = \underbrace{\sum_{x: x<a} x\,\mathbb{P}(X=x)}_{\geq 0 \text{ since } X:\text{nonneg}} + \sum_{x: x \geq a} x\,\mathbb{P}(X=x) \geq \sum_{x: x \geq a} x\,\mathbb{P}(X=x) \geq \sum_{x: x \geq a} a\,\mathbb{P}(X=x) \overset{\text{additivity}}{=} a\,\mathbb{P}(X \geq a)$ $\square$

REMARK

Markov's ineq gives us "small expectation $\Rightarrow$ random variable is small whp"

HOWEVER, large expectation $\not\Rightarrow$ random variable high whp

Example:

$$X_n = \begin{cases} n^2 & \text{w/p } \frac{1}{n} \\ 0 & \text{w/p } 1-\frac{1}{n} \end{cases} \Rightarrow \begin{array}{l} \mathbb{E}[X_n] = n^2 \cdot \frac{1}{n} = n \xrightarrow{n\to\infty} \infty \\ \mathbb{P}(X_n \neq 0) = \frac{1}{n} \xrightarrow{n\to\infty} 0 \end{array}$$

CHEBYSHEV'S INEQUALITY (SMALL VARIANCE $\Rightarrow$ CLOSE TO EXPECTATION WHP)

Let X be a random variable w/ $\mu = \mathbb{E}[X]$ on a probability space $(S, \mathbb{P})$, and let $a > 0$ be a real number.

Then, $\mathbb{P}(|X-\mu| \geq a) = \frac{Var(X)}{a^2}$

Proof

Apply Markov to $Y = (X-\mu)^2 \geq 0$, then $\mathbb{P}(|X-\mu| \geq a) = \mathbb{P}(Y \geq a^2) \leq \frac{\mathbb{E}[Y]}{a^2} = \frac{\mathbb{E}[(X-\mu)^2]}{a^2} = \frac{Var(X)}{a^2}$ $\square$

COROLLARY

The probability that X is more than k std deviations from its expectation is $n \leq \frac{1}{k^2}$

Proof

Apply Chebyshev with $a = k\sigma(X)$

Then, $\mathbb{P}(|X-\mu| > k\sigma(X)) \leq \underbrace{\frac{Var(X)}{(k\sigma(X))^2}}_{\text{by def}} = \frac{1}{k^2}$

OBSERVE

Let X be a random variable, $a, b \in \mathbb{R}$.

Then, $\mathbb{E}[aX+b] = a\mathbb{E}[X] + b$ by linearity

$Var(aX+b) = \mathbb{E}[((aX+b) - \mathbb{E}[aX+b])^2] = \mathbb{E}[((aX+b) - (a\mathbb{E}[X]+b))^2] = a^2\mathbb{E}[(X-\mathbb{E}(X))^2] = a^2 Var(X)$

DISCRETE DISTRIBUTIONS

BERNOULLI DISTRIBUTION

Denoted via $Ber(p)$ where $0 \leq p \leq 1$

If $\underbrace{X \sim Ber(p)}_{X \text{ has the } Ber(p) \text{ distribution}}$, then

$$X = \begin{cases} 1 & \text{w/p } p \\ 0 & \text{w/p } 1-p \end{cases}$$

We get the following results:

↳ $\mathbb{E}[X] = 1 \cdot p + 0(1-p) = p$

↳ $Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X] - \mathbb{E}[X]^2 = p - p^2 = p(1-p)$

BINOMIAL DISTRIBUTION

Denoted by $Bin(n,p)$, $n \in \mathbb{N}$, $0 \leq p \leq 1$

We run n indep trials, each with success probability p. $Bin(n,p) = \#\text{successes}$

↳ Possible values: $0, 1, \ldots, n-1, n$

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EXAMPLES OF BINOMIAL DISTRIBUTIONS

- # heads in n coin tosses
- # odd rolls in n rolls of a die
- # errors when replicating DNA
- # supporters of a particular party in a poll

IMPORTANT QUANTITIES

Prob mass function: $\mathbb{P}(X=k) = \underbrace{\binom{n}{k}}_{\text{choose which trials succeed}} \underbrace{p^k}_{k \text{ successful trials}} \underbrace{(1-p)^{n-k}}_{n-k \text{ unsuccessful trials}}$

For $\mathbb{E}(X)$,

Method 1: $\mathbb{E}(X) = \sum_{k=0}^{n} k\mathbb{P}(X=k) = \sum_{k=0}^{n} k\binom{n}{k} p^k (1-p)^k$

Method 2: $X = \sum_{i=1}^{n} X_i$, where $X_i := \begin{cases} 1, & \text{trial } i \text{ accepted w/p } p \\ 0, & \text{o/w w/p } 1-p \end{cases}$

As each $X_i \sim Ber(p)$, thus $\mathbb{E}[X] = \mathbb{E}[\sum_{i=1}^{n} X_i] \overset{\text{linearity}}{=} \sum_{i=1}^{n} \mathbb{E}[X_i] \overset{X_i \sim Ber(p)}{=} \sum_{i=1}^{n} p = np$

Method 3

Recall the Binomial Thm, $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$

Differentiate w.r.t. x, $\frac{\partial}{\partial x}((x+y)^n) = \frac{\partial}{\partial x}\left(\sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}\right) \Rightarrow n(x+y)^{n-1} = \sum_{k=0}^{n} k\binom{n}{k} x^{k-1} y^{n-k}$

Multiplying by x, we get $nx(x+y)^{n-1} = \sum_{k=0}^{n} k\binom{n}{k} x^k y^{n-k}$ (☆)

Sub $x=p$, $y=1-p$, $np(p+(1-p))^{n-1} = \sum_{k=0}^{n} k\binom{n}{k} p^k (1-p)^{n-k} \Rightarrow \mathbb{E}(X) = np$

For $Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, $\mathbb{E}[X^2] = \sum_{k=0}^{n} k^2 \binom{n}{k} p^k (1-p)^{n-k}$

Diff (☆) w.r.t. x, $n(x+y)^{n-1} + n(n-1)x(x+y)^{n-2} = \sum_{k=0}^{n} k^2 \binom{n}{k} x^{k-1} y^{n-k}$

Multiplying by x, $nx(x+y)^{n-1} + n(n-1)x^2(x+y)^{n-2} = \sum_{k=0}^{n} k^2 \binom{n}{k} x^k y^{n-k}$

Sub $x=p$, $y=1-p$, $np + n(n-1)p^2 = \sum_{k=0}^{n} k^2 \binom{n}{k} x^k y^{n-k} = \mathbb{E}[X^2]$

$\therefore Var(X) = np + n(n-1)p^2 - (np)^2 = np - np^2 = \underbrace{np(1-p)}_{= nVar(Ber(p))}$

POISSON DISTRIBUTION

Denoted as $Poi(\lambda)$, where $\lambda \geq 0$ is a real number

Setting: Rare indep events, count #occurances in a given time period, where $\lambda :=$ expected # occurances

No pattern in occurance

Possible values of $X \sim Poi(\lambda)$: $0, 1, 2, 3, \ldots$

EXAMPLES OF POISSON DISTRIBUTIONS

- Earthquakes
- # Customers entering a store
- # Phone calls received